Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/24/2025
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Correlation of Normal Gravity Aerospace Materials Flammability Data with Flammability in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	216-433-2859
Organization Name:	NASA Glenn Research Center		
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City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	09/30/2027
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:			
Flight Assignment:	NOTE: Changed end date to 09/30/2027, per PI (Ed., 3/7/25) NOTE: Changed end date to 12/31/2024, per PI (Ed., 11/28/22) NOTE: Changed end date to 12/31/2022, per PI (Ed., 10/3/19) NOTE: Extended to 12/31/2019, per PI (Ed., 1/26/2015) NOTE: Extended to 12/31/2014, per PI (Ed., 10/17/13) NOTE: Extended to 12/31/2013, per PI (Ed., 10/10/2012)		
Key Personnel Changes/Previous PI:	March 2025 update: Per the Principal Investigator (PI), David Hirsch is no longer with the project, so he has been removed from the list of Co-Investigators (Ed., 3/7/25).		
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	NASA-STD-6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is an upward flame propagation test initiated in a stagnant environment and using a well-defined igniter flame at the bottom of a vertically mounted sample. A material passes this test if the vertical burn length is less than 15.2 cm (6 inches) and there is no evidence of transfer of burning debris. The test is conducted in the most severe flaming combustion environment (oxygen concentration, pressure) expected in the spacecraft, currently 30% oxygen at 10.2 pounds per square inch absolute (psia). Recent research has shown that current normal gravity materials flammability tests do not correlate with flammability in ventilated, micro-, or reduced-gravity conditions. The materials selection for spacecraft is based on the assumption of commonality between ground test flammability results and spacecraft environments, which does not appear to be valid. To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just "passing" from a flammability point of view. A research program is underway to correlate normal gravity flammability test data with data under ventilated reduced gravity or microgravity conditions. The program will determine materials suitability for microgravity and reduced gravity testing by evaluating oxygen concentration thresholds under various total pressures in normal gravity tests and in spacecraft-specific environments (ventilated microgravity or reduced partial gravity). The normal gravity data will be correlated with flammability data in microgravity environments and the safety factors involved when selecting spacecraft materials based on normal gravity testing will be evaluated. Rigorous correlations between ground test flammability data and data in ventilated spacecraft environments would allow selection of materials with increased fire resistance, and thus decrease the fire risk in space systems. The operational flexibility will also increase since the safety factors will be assessable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of the fundamental limit in microgravity and its comparison with the normal gravity flammability limit will help us understand flammability in different configurations on Earth. It will help us understand the impact of buoyant flow on flammability, and help in evaluating flammability on the Moon or Mars. It will improve the safety of astronauts during exploration missions so the mission can successfully be completed and the crew can return safely to Earth.
Task Progress:	During the past year, SoFIE (Solid Fuel Ignition and Extinction) GEL (Growth and Extinction Limits), MIST (Material Ignition and Spread Test), and RTDFS (Residence Time-limited Diffusion Flame Spread) have been underway, and I am part of the SoFIE science team. During these operations, I have become very familiar with the capabilities and limitations of the SoFIE hardware, which is very useful knowledge for planning to maximize the science return for SMµRF. The current Artemis mission vehicle pressure is approximately 56 kPa, so microgravity flammability tests at this pressure are of significant interest for future exploration missions. Testing at this pressure was conducted in the NASA Zero Gravity Research Facility Centrifuge drop vehicle at various gravity levels up to simulated normal gravity [Fire Safety Journal, 2024]. The flammability limits for PMMA rods at the exploration pressure and different gravity levels were found as shown in the figure. The most flammable condition (the lowest oxygen that supports a flame) appears to be near lunar gravity levels. This suggests fire safety on the Moon is a very serious risk that needs careful attention. The SMµRF experiment operations are currently scheduled to start in the spring of 2026. Flat samples will be tested first, followed by rod samples. In addition to flammability limit evaluation, data analysis methods are being developed to analyze flame spread, surface regression, smolder, and bubble layer behaviors.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Olson SL. "Characterizing the wave of discrete glowing smolder spots generated at the base of a concurrent microgravity flame spreading over a cotton-fiberglass fabric." Proceedings of the Combustion Institute. 2024;40(1-4):105495. https://doi.org/10.1016/j.proci.2024.105495 , Jul-2024
Articles in Peer-reviewed Journals	Olson SL, Neumann ES, Valletta VM, Ogorzaly AL, Owens JC. "Partial gravity flammability of cast PMMA rods in concurrent axial stagnation flow." Fire Safety Journal. 2024 Dec 1;150:104267. https://doi.org/10.1016/j.firesaf.2024.104267 , Dec-2024